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Research Assignment 1

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Section A: Data Fundamentals

① **Database**: An organized collection of structured (data in an organized & fixed format such as a table with rows & columns) information or data, typically stored electronically in a computer system. Examples include a Raw Material Database which stores information about ingredients and suppliers, as well as a Microbiology Database which stores microbiological testing results.

② A **data warehouse** is a central system that stores cleaned and organized historical data pulled from many source systems. A data warehouse differs from a data base in the sense that it collects information from ~~del~~ different data bases in order to conduct business analysis whereas a database is for daily operations. Essentially the database tells us what happened and the data warehouse helps us understand what keeps happening & why.

③ A **data lake** is a centralized repository that stores large volumes of raw data ~~it~~ in its original format. It can contain structured data (tables and databases), semi-structured data (JSON & XML), and unstructured data (documents, images, videos and emails). Data is stored first and structured when needed. On the contrary, a **data warehouse** stores cleaned, structured (tables & databases) and transformed data.

Data Lake - Schema-on-read; Store everything first, decide how to use it later.

Data Warehouse - Schema-on-write; Organise important data first, then use it for reporting.

A business would prefer to use a data lake when dealing with dealing with diverse or unstructured data such as Production Records, Product photos, Temperature and humidity readings, Supplier COAs.

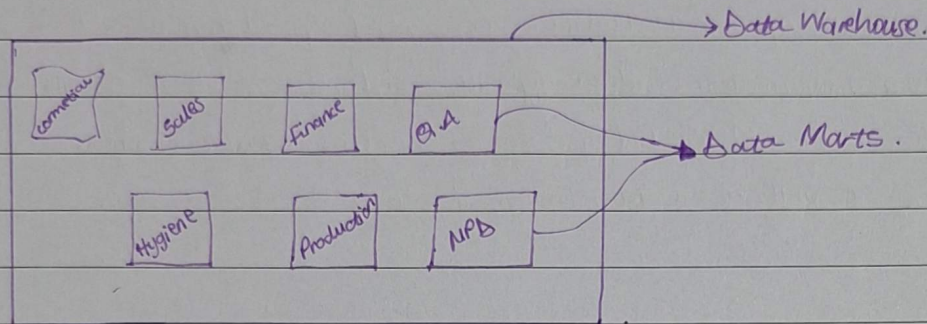
④ A **data mart** is a smaller, focused section of a data warehouse designed for the needs of a specific department or business function. An organization can have one central Data Warehouse containing data from the entire organization. Data Marts can be created to include the below:

① Quality Assurance: NCRs, Complaints, Audits & Supplier Performance.

② Finance - Budgets, Costs, Profitability.

③ Sales - Sales Performance, Customers and Revenue.

~ So like a Mall (Database) and all the different stores (Data Marts)



⑤ **Structured Data** is data that is highly organised and follows a predefined format such as a table with rows and columns. *A customer database that contains names, customer IDs & contact details.*

- **Semi-structured Data** is data that does not follow a strict tabular structure but contains labels, tags or other organisational elements that make it easier to process. *A JSON File.*

- **Unstructured Data** is data that has no predefined format or organisation and is more difficult to store & analyse using traditional databases. *A video file or photograph.*

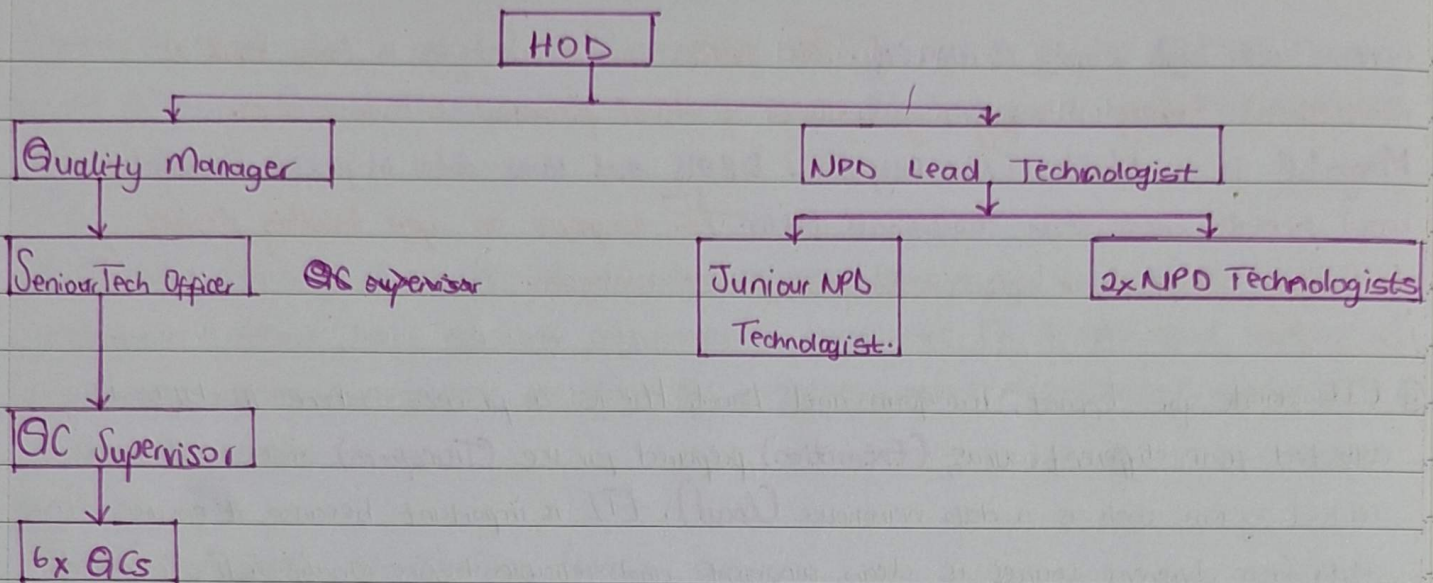
⑥ **Data Modelling** is the process of designing how data will be organized, stored, connected & managed within a system. It is important because it provides a blueprint of how different pieces of data relate to one another and also reduces duplicate/unnecessary data elements.

Two types of Data Models

① **Relational Data Model:** A company can have separate (DIM) tables consisting of rows & columns, with relationships between tables. For example having separate tables for customers, orders and products and connected using customer/product IDs (DIM Tables).

② **Hierarchical Data Model:** Organises data in a tree-like structure where each item can have a parent and one or more children. e.g. Organograms

Company → Departments → Employees → Employee Records



⑦ Data Mining is the process of analyzing large sets of data to discover useful patterns, trends, relationships and hidden insights. Businesses use data mining to improve marketing strategies, identify customer buying patterns and detect fraud / unusual activity. Some real world examples include: banks being able to analyse transaction data to identify unusual patterns such as a customer making a large purchase in a different country shortly after making a local one. Secondly a food manufacturing company can analyse historical sales data to identify patterns in customer demand. eg data mining can reveal that sales of chocolate & confectionery products increase significantly during easter & christmas. The business can then use this information to plan production, purchase raw materials & manage stock levels.

⑧ Database Management System (DBMS) is a software that allows users and organisations to create, store, organize, retrieve, and manage data in a database. It assists in ensuring that data is stored efficiently, securely and can be accessed when needed.

MySQL is a popular open-source relational DBMS that stores data in tables with rows & columns. This DBMS can be used by a small-to-medium sized business website to manage customers, products & orders.

PostgreSQL is an advanced open source relational DBMS known for reliability & support for complex queries and data types. A business system requiring complex data analysis, such as a financial or quality management system.

Oracle Database is a powerful commercial relational DBMS designed for large-scale, enterprise level

systems with high security & reliability. This DBMS can be used by a large bank or multinational company managing large volumes of critical financial & customer data.

MongoDB is a NoSQL (Not Only SQL) DBMS that stores data in flexible, document-based formats rather than traditional tables. For example an app handling flexible or changing data, such as social media posts, product catalogues.

9) ETL stands for Extract, Transform and Load. It is a process where data is collected from different sources, (Extraction) prepared for use (Transform) and stored in a central system such as a data warehouse (Load). ETL is important because it ensures that data from different sources is clean, consistent and reliable before analysis. This improves data quality and allows businesses to combine information, identify trends and make more accurate decisions.

10) CSV (Comma-Separated Values)

This format is used to store data in simple rows-and-columns format, similar to a spreadsheet. It is commonly used when one needs a simple format for sharing/transferring structured data between systems. e.g. Exporting sales data from Excel. Simple tables

JSON (JavaScript Object Notation)

This format stores data using key-value pairs making suitable for structured and semi-structured data. It can be used when working with web applications. e.g. product information received from an online API. Flexible Records

Parquet

This is a column-based storage format designed for efficient storage and analysis of large datasets. It is used when working large datasets and data lakes. e.g. analysing millions of customer transactions. Efficient large-scale analytics.

Avro

This is a compact, row-based format that stores data together with its schema (the structure describing the data). It is useful when data needs to be efficiently transferred between systems, particularly in big-data and streaming environments. Efficient data-transfer/streaming.

Section B: Artificial Intelligence & Machine Learning.

11) Artificial Intelligence (AI) is the ability of machines or computer systems to perform tasks that normally require human intelligence, such as learning, reasoning & problem-solving.

Narrow AI is AI that is designed to perform a specific task such as Siri or facial recognition system. Whereas General AI speaks to a theoretical AI that could think, learn & perform any intellectual task like a human.

(12) Machine Learning is a type of AI where computers learn from data and improve their performance without being explicitly programmed for every task. On the other hand traditional programming involves a programmer providing the computer with specific rules & data to produce an output.

The 3 main types of Machine Learning are:

Supervised Learning which uses labelled data to train a model to make predictions. For example it can be used to identify whether an email is spam or not.

Unsupervised Learning uses unlabelled data to identify hidden patterns or groups. For example it can group customers according to their purchasing behaviour.

Reinforcement Learning allows a system to learn through rewards & penalties based on its actions. For example, it can be used to train a robot to perform tasks.

(13) Deep Learning is a subset of Machine Learning that uses complex Neural Networks with multiple layers to learn patterns from large amounts of data. Neural Networks are computer systems inspired by the structure of the human brain. They consist of connected layers of artificial neurons:

- Input layer which receives the data.
- Hidden layers which process the data & identify patterns.
- Output layer which produces the final result/prediction.

(14) Natural Language Processing (NLP) is a branch of AI that enables computers to understand, interpret & generate human language. For example Chat GPT understanding questions & giving human-like responses.

(15) Large Language Model (LLM) is an AI model trained on large amounts of text to understand and generate human-like language. Unlike a standard machine learning model, which is usually designed for a specific task, an LLM can perform many language related tasks such as summarising, translating, generating texts & answering questions.

Two well known LLMs:

- > ChatGPT for conversations, answering questions, summarising, writing etc.
- > Google Gemini for answering questions, generating content, analysing information etc.

⑥ Generative AI is a type of AI that can create new content, such as text, images, audio, video or code, based on patterns learned from existing data. At a high level it works by learning patterns from large amounts of data and then using those patterns to generate new content in response to user instruction / prompt.

Examples:

- ① DALL-E generates ~~text~~ images from text descriptions
- ② GitHub Copilot generates and suggests computer code based on the developer's instruction.
- ③ ChatGPT generates text, answers questions, summarize information and assists with tasks.

⑦ Training dataset is a collection of data used to teach an AI model to recognize patterns and make predictions / decisions. The quality of the training data is very important because AI models learn from the data they are given. Accurate, complete and representative data helps produce reliable results. If the training data is biased, incomplete, or incorrect, the AI model may learn those errors and produce biased, inaccurate or unfair results.

⑧ Computer vision is a branch of AI that enables computers to interpret and understand images and videos. Machines 'see' by using cameras or image files to capture visual information. They analyse patterns such as shapes, colours, objects & faces and interpret.

Real-world uses

- ① Facial Recognition: Computer vision can identify / verify people using their facial features
- ② Manufacturing Quality Control: Cameras can detect defects, damaged products or incorrect packaging on the production line.

Section C: APIs, Tools & Developer Concepts

⑨ API Application Programming Interface

An API is a set of rules that allows different software applications to communicate and exchange data. An API is like a waiter the QC in a food manufacturing facility. You (the QA Manager) send the QC to the floor to request specific feedback from the Production Manager. The QC goes to the

floor and gets this feedback and brings it back to you. e.g. the weather app.

20 An API key is a unique code used to identify and authenticate an application when it accesses the API. Its purpose is to control access, track usage and help prevent unauthorized use of an API. If an API key is exposed, someone could use it without permission, potentially causing unauthorized access, data breaches / unexpected charges.

21 A Rest API is an API that allows applications to communicate and exchange data over the internet using standard HTTP methods. **REST** stands for **Representational State Transfer**. The Main HTTP methods are:

- ① GET: Retrieves or reads data.
- ② POST: Creates or adds new data.
- ③ PUT: Updates existing data.
- ④ DELETE: Removes data.

22 JSON (JavaScript Object Notation) is a simple, text-based format used to store and exchange data. It organizes information using key-value pairs. JSON is commonly used with API because it is easy for humans to read and easy for computer systems to process & exchange.

</> JSON

```
{
  "product": "Caramel Fudge",
  "weight": "100g",
  "price": 35
}
```

In the above example "product", "weight" and "price" are the keys, while their associated information are the values

23 Prompt Engineering is the skill of creating clear and effective instruction for AI tools to produce accurate and useful responses. It is important because a clear, specific prompt gives AI enough context to provide a better result.

Example 1

Poor Prompt: Write about Food Safety

Improved Prompt: Write a 200 word explanation of food safety for food manufacturing employees using very simple language.

Example 2

Poor Prompt: Write me a professional email.

Improved Prompt: Write a professional email to Relianz requesting an updated COA for the raw materials. Keep the tone polite and concise.

②4 The MAP function applies the same operation or function to every item in a collection such as a list, and returns a new collection with the results.

`</> Python`

```
numbers = [1, 2, 3, 4]
```

```
results = list(map(lambda x: x*2, numbers))
```

```
print(results)
```

Output

```
[2, 4, 6, 8]
```

②5 Cloud Computing is the delivery of computing services such as storage, servers, software and databases over the internet, instead of using only local computers or servers.

① IaaS (Infrastructure as a Service) provides virtual servers, storage and networking that users can manage. E.g Amazon EC2

② PaaS (Platform as a Service) provides a platform and tools for developers to build and run applications without managing the underlying infrastructure. E.g Google App Engine.

③ SaaS (Software as a Service) provides ready-to-use software over the internet. E.g Microsoft 365.

②6 The Version Control System (VCS) is a tool that tracks changes to files over time, allowing users to view previous versions and collaborate without losing work. GIT is a popular version control system widely used by developers and data analysts to track code, collaborate with others, manage changes and restore previous versions.

A 'commit' is a saved record of changes made to a project. Whereas 'branch' is a separate version of a project where changes can be made without affecting the main version.

27) ~~What is~~ A Jupyter Notebook is an interactive tool that allows users to write and run code, analyse data, and display results such as tables and graphs in one document. It is most commonly used with Python. Jupyter Notebooks are popular among Data Analysts & Data Scientists because they make it easy to analyse data step-by-step, visualize results, test codes and document the analysis in one place.

28) Python is a high-level programming language known for its simple and readable syntax. It is popular in Data Analytics and AI because it is easy to learn, flexible and has many powerful libraries for working with data and machine learning.

Four commonly used Python libraries are:

- 1) Pandas - Used for cleaning, organizing and analysing data.
- 2) NumPy - Used for numerical calculations and working with arrays.
- 3) Matplotlib - Used to create graphs and visualisations.
- 4) Scikit-learn - Used to build and evaluate Machine Learning models.

29) Section D: Data Ethics, Governance & Careers.

30) Data Privacy is the protection of personal information and ensuring that it is collected, stored, used and shared responsibly. It is important because it protects individuals from unauthorized access, misuse, identity theft and data breaches.

GDPR - General Data Protection Regulation is a European Union Law that protects individuals' personal data & regulates how organisations collect and process it. It has applied since 25 May 2018. The GDPR applies to all the EU countries and through the EEA agreement, also applies in Iceland, Liechtenstein and Norway.

31) Data Ethics refers to the principles that guide how data and AI should be collected, used and managed fairly, responsibly and transparently.

3 examples of unethical uses are:

1. Using personal data without consent: Collecting/Selling people's personal information without their knowledge violates privacy & can lead to misuse of their data.

② **Biased AI decision-making**: Using an AI system that unfairly favours or disadvantages certain groups can lead to discrimination and unfair decisions.

③ **Using AI to create fake information**: Generating fake images, videos or news to deceive people can spread misinformation and damage individuals or organisations.

④ **Data Governance** is the framework of policies, rules and processes used to ensure that an organisation's data is accurate, secure, accessible and used responsibly.

Key components include:

① **Data Quality**: Ensure data is accurate, complete and consistent.

② **Data Security**: Protects data from unauthorized access, loss or misuse.

③ **Data Privacy**: Ensures personal information is handled appropriately and legally.

④ **Data Policies & Standards**: Establishes rules for how data should be collected, stored and used.

⑤ **Roles and Responsibilities**: Defines who is responsible for managing and accessing data.

⑥ **Compliance**: Ensures the organisation follows relevant laws and regulations.

⑤ **AI bias** occurs when an AI system produces unfair or systematically inaccurate results because of biases in its training data, design or how it is used. For example, some facial recognition systems have been found to be less accurate when identifying people from certain demographic groups. As a potential consequence this can lead to wrong identifications, discrimination & unfair decisions, particularly when AI is used in areas such as security, employment or law enforcement.

DATA ROLES

③

Role	Main Responsibility	Key Skills	Common Tools
Data Analyst	Analyses data to identify trends and support business decisions.	Excel, SQL, Statistics, data visualisation.	Excel, SQL, Power BI, Tableau.
Data Scientist	Uses data, statistics and machine learning to make predictions and solve complex problems.	Python, statistics, machine learning.	Python, pandas, Scikit-learn, Jupyter.

Delta Engineer.	Builds and maintains systems that collect, store and process data.	Data bases, programming, data pipelines, cloud computing.	SQL, Python, Spark, AWS/Azure.
AI Engineer	Develops and deploys AI and machine learning systems.	Machines learning, programming, AI models, APIs.	Python, TensorFlow, PyTorch, APIs.

34) Business Intelligence (BI) is the use of data, tools and processes to analyse business information and support better decision-making.

Two Popular BI tools are:

- ① Power BI - creates interactive dashboards and reports that help businesses monitor performance and identify trends.
- ② Tableau - Converts data into visualisations and dashboards, helping users understand patterns and make informed decisions.

Section E: PERSONAL REFLECTION

35) What I found very interesting is connected everything really is. As an individual who is in the food manufacturing space the amount of examples I was able to give based off of my profession was very eye opening. The section that I found most interesting about the assessment was definitely Section D on Data Ethics, Governance & Careers - again linking to my regulatory compliance in the food industry the protection of intellectual property, including data, across industries.

36) I found section C on APIs, Tools & Developer Concepts quite challenging. To help with my understanding I definitely had my fair share of Youtube videos. Although there is still much to learn. There is light.

37) I definitely see an improvement in how I handle everyday task. I am currently building and implementing a Food Safety and Management System. There is A LOT of data that needs to be collected. From a Supplier Quality Assurance System for both packaging & raw materials to keep track of production records to customer complaints to laboratory analysis results and trending.

References

1. Class Notes
2. Google Gemini
3. [Cloud Data Lake vs. Data Warehouse vs. Data Mart | IBM](#)
4. https://www.ibm.com/think/topics/datamart?a=PXD6EMQD&utm_source=chatgpt.com
5. https://www.ibm.com/think/topics/data?utm_source=chatgpt.com
6. https://www.ibm.com/think/topics?utm_source=chatgpt.com
7. https://www.ibm.com/think/topics/ai-computing?utm_source=chatgpt.com
8. https://www.nist.gov/itl/ai-risk-management-framework?utm_source=chatgpt.com
9. https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1673360464141&uri=CELEX:32016R0679&utm_source=chatgpt.com
10. https://www.nist.gov/itl/ai-risk-management-framework/ai-risk-management-framework-resources?utm_source=chatgpt.com
11. https://docs.python.org/3/?utm_source=chatgpt.com
12. [Cloud Data Lake vs. Data Warehouse vs. Data Mart | IBM](#)
13. [Power BI Tutorial for Beginners](#)
14. [Tableau Basics for Beginners - Tableau in Two Minutes](#)